

ACG Architecture & System Design

Merchant-Side Control Plane for AI-Originated Transactions

1. System Pipeline & Core Boundaries

The Agent Commerce Gateway operates as an explicit, 6-phase zero-trust pipeline. No financial action can proceed directly from an AI model to a payment rail without passing all authorization gates.

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ANY AI MODEL / AGENT (Proposer)
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▼
PROTOCOL / AGENT ADAPTER (Normalizer)
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▼
CANONICAL FINANCIAL INTENT (Intermediate Representation)
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▼
MANDATE VERIFICATION (Ed25519 & Revocation Registry)
|
▼
MERCHANT TRUTH (Authoritative Catalog Price & Stock)
|
▼
MERCHANT POLICY (Versioned Policy DSL Evaluation)
|
▼
ATOMIC DUAL-RESOURCE RESERVATION (Budget Paise + Stock Units)
|
▼
AUTHORIZATION DECISION (Permit / Block with Deterministic Code)
|
▼
RAZORPAY EXECUTION (Order Creation with receipt = intent_id)
|
▼
WEBHOOK RECONCILIATION (HMAC SHA-256 & Event Deduplication)
|
▼
TAMPER-EVIDENT SHA-256 AUDIT LEDGER (Provenance & Verification)
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2. Separation of Concerns

- AI Proposer (Zero Authority):** AI agents, LLMs, and subagents synthesize buyer intent and propose items, quantities, and client nonces. They possess **zero authority** to set prices, modify taxes, or self-authorize execution.
- ACG Control Plane (Deterministic Authorizer):** ACG inspects cryptographic delegation, validates against merchant database truth, applies active merchant policy, and acquires ACID locks across budget and stock.
- Razorpay Rails (Financial Executor):** Razorpay receives pre-authorized, idempotent orders, captures customer funds, executes refunds, and dispatches HMAC-signed webhook notifications.

3. Financial Action Types

ACG models explicit financial action types:

* **AUTHORIZE_AND_RESERVE:** Evaluates intent against policy and locks budget and inventory.

- * **CREATE_PAYMENT_ORDER**: Invokes Razorpay POST /v1/orders with receipt = intent_id.
- * **CAPTURE_PAYMENT**: Processes Razorpay payment.captured webhook and commits dual-resource locks.
- * **RELEASE_RESERVATION**: Aborts and rolls back reserved budget and inventory upon timeout, cancellation, or rail failure.
- * **EXECUTE_REFUND**: Issues idempotent refund via X-Refund-Idempotency when post-capture fulfillment fails.

4. State Transition Machine

[illegible]